- `fopen()` returns a file pointer (`FILE*`) that refers to the file, which can then be used for subsequent file operations. If the file cannot be opened, `fopen()` returns `NULL` .

File Opening Modes: The mode parameter in the `fopen()` function determines how the file will be accessed. The common modes include:

- `"r"` : Opens a file for reading. If the file doesn't exist, an error is returned.
- `"w"` : Opens a file for writing. If the file exists, its contents are cleared; if it doesn't exist, a new file is created.
- `"a"` : Opens a file for appending. Data is added to the end of the file. If the file doesn't exist, it's created.
- `"r+"` : Opens a file for both reading and writing. The file must exist.
- `"w+"` : Opens a file for both reading and writing. The file is created if it doesn't exist; existing contents are cleared.
- `"a+"` : Opens a file for reading and appending. Data is written at the end of the file.

Example of opening a file for reading:

```
FILE *file = fopen("data.txt", "r");

if (file == NULL) {

    printf("Error opening file.\n");

    exit(1);  // Exit if file cannot be opened

}
```